Lab 01- Marks 15: (Weightage- 1.5%)

Exercise 1: User Input and Printing

1. Ask the user to enter their name and print a greeting message.
2. Ask the user to enter their age and print it.
3. Ask the user to enter their favorite color and print a summary of the information.

Exercise 2: Checking Types and IDs

1. Create variables:

   a = 10

   b = 3.5

   c = "Python"

   d = True

2. Print the type of each variable using type().
3. Print the memory address of each variable using id().
4. Compare two variables with the same value (e.g., x = 100, y = 100) and check if their IDs are the same.

Exercise 3: Using eval()

1. Ask the user to enter a mathematical expression like 2 + 3 * 4.
2. Use eval() to calculate the result and print it.
3. Ask the user to enter a list of numbers as a string (e.g., [1,2,3,4]) and convert it to a list using eval().
4. Print the type of the result after using eval().

Exercise 4: Exploring Objects with dir() and help()

1. Create a string variable text = "Python". Use dir() to list all its methods and attributes and print the result.

2. Create a list variable numbers = [1,2,3] and print its dir() output.
3. Ask the user to input the name of a Python function (like len or print) and use help() to display its documentation.
4. Explain how dir() and help() can help you understand Python objects and functions.

Exercise 5: Combined Practice

1. Ask the user to enter two numbers as strings and use eval() to convert them to numbers.
2. Calculate and print the sum, difference, and product of the numbers.
3. Check and print the type and memory address of each number.
4. Ask the user to enter a Python expression of their choice, evaluate it using eval(), and print both the result and its type.